

Phase 2 (Parity + Reach) Subtask Deepening — epic → task → subtask (closes R5)

Phase 2 (Parity + Reach) Subtask Deepening — epic → task → subtask (closes R5)

Revision: 1 Last modified: 2026-06-26T12:00:00Z

Volume 7 (Phase Execution), document 4 of 5. Companion to subtask-deepening-p1.md, applying the same R5 deepening (REFINEMENT_NOTES.md) to Phase 2: every task HVPN-P2-NNN in 08-phase2-parity-wbs.md is decomposed into PR-sized subtasks HVPN-P2-NNN.k (§11.4.93/.54), each with a concrete **acceptance** (falsifiable, captured-evidence §11.4.5/.69/.107), the **§11.4.169 test types** (the 08-... §1 vocabulary, now including DDOS — re-armed for the public multi-region surface), and an **estimated complexity** (XS/S/M/L, sizing-only TARGET, §11.4.6). **Spec-only.** Phase 2's load-bearing premise — *everything is additive* (08-... §0) — means most subtasks are "more impls of an existing seam", and two epics (E21 DAITA, E24 PQ) carry the *measure-don't-assert* privacy-claim discipline (self-validated analyzers, §11.4.107(10)). Every subtask traces to its parent task in 08-...; nothing is invented beyond decomposing stated work. These .k rows feed the workable-items DB (workable-items-model.md §7).

Table of contents

- 0. Deepening conventions
- 1. E19 — Phase-1 seam certification (entry)
- 2. E20 — Full transport / obfuscation set
- 3. E21 — DAITA traffic-analysis defense
- 4. E22 — Direct P2P + NAT traversal + relay
- 5. E23 — Multi-hop (nested WireGuard)
- 6. E24 — Post-quantum handshake (ML-KEM PSK)
- 7. E25 — Desktop apps (Windows, macOS)
- 8. E26 — Policy-as-code & GitOps
- 9. E27 — HA + multi-region fleet
- 10. E28 — Observability additions

- 11. E29 — QA · SLOs · DoD certification
- 12. E30 — Governance & release
- 13. Subtask roll-up + measure-don't-assert note
- Sources verified

0. Deepening conventions

Identical to subtask-deepening-p1.md §0: id (HVPN-P2-NNN.k) · **Subtask** (≥ 6 words §11.4.91) · **Acceptance** (falsifiable, captured-evidence) · **Tests** (§11.4.169 codes, 08-... §1 vocabulary incl. DDOS) · **Cx** (XS/S/M/L sizing TARGET). A subtask is complete only with a test_diary evidence path (workable-items-model.md §9). **Two anti-bluff escalations specific to Phase 2** (08-... §0): (a) any "direct path" claim is Direct only when a non-gateway src is observed at the peer (§11.4.107 — no false direct); (b) DAITA + PQ privacy claims **MUST** be *measured* with a self-validated analyzer (golden-bad fixture must FAIL the analyzer), never asserted. Complexity sums to $\approx$ parent effort (08-... §19, ~ 259 person-days excl. entry seams) — sizing only, never a date.

1. E19 — Phase-1 seam certification (entry)

Entry gate (not Phase-2 work; tracks that the reserved seams exist, 08-... §5).

HVPN-P2-190 — Transport-trait seam present (XS) · 191 — Peer.endpoint candidate-list seam (XS) · 192 — federated Coordinator + PKI hooks (XS).

Each is a single seam-audit subtask:

id	Subtask	Acceptance	Tests	Cx
190.1	Register a no-op stub transport + walk the ladder unchanged	the ladder walks the stub with zero changes to the trait	UNIT,INT	XS
191.1	Audit NetworkMap.Peer.endpoint is a repeated EndpointCandidate	the field is repeated; a coordinator delta can populate it	UNIT	XS
192.1	Audit coordinator statelessness + a PKI PSK-injection hook	coordinator hydrates from store+events; PKI exposes a PSK hook	UNIT,SEC	XS

2. E20 — Full transport / obfuscation set

Every new transport is just another impl of the unchanged Phase-1 Transport trait.

HVPN-P2-200 — Shadowsocks-wrap transport (08-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Wrap WG datagrams in a Shadowsocks AEAD stream (reuse shadowsocks-rust primitives)	tunnel survives udp drop+quic drop nft rules (E2E)	UNIT,INT,E2E	M
.2	DPI-gauntlet classifier check	tshark sees no WG/QUIC signature (SEC)	SEC,CHAL	S
.3	Goodput record with TCP-HoL caveat	goodput captured + HoL noted (BENCH)	BENCH	S

HVPN-P2-201 — UDP-over-TCP (UoT) transport (08-... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	Length-prefix WG over a single TCP conn; ladder ranks last	tunnel comes up under udp drop (all ports); reachability preserved (E2E)	UNIT,E2E	S
.2	HoL penalty measured + documented	HoL overhead captured (BENCH)	BENCH	XS

HVPN-P2-202 — LWO hardening (per-session keyed header obfuscation) (08-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Per-session-keyed obfuscation of WG header signature bytes + randomized padding	a WG-signature DPI rule that drops plain-WG fails to classify LWO-v2 (SEC)	UNIT,SEC,CHAL	M
.2	Overhead < 2% vs plain-UDP	captured overhead < 2% (BENCH)	BENCH	XS

HVPN-P2-203 — Refined auto-escalation ladder + per-network memory (08-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Ladder FSM with per-step failure budget (N handshakes/T seconds)	client converges to shadowsocks on a UDP+QUIC-blocked network (E2E)	UNIT,E2E	M
.2	network_memory per SSID/gateway fingerprint	on the next connect to the same network, escalation latency is skipped (captured timing delta)	FA,CHAL	M

HVPN-P2-204 — Coordinator-pushed regional priors (08-... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	TransportPolicy.region_prior map in NetworkMap; region-keyed reorder	a censored-region-resolved client gets a reordered ladder + connects on the prior's head transport	UNIT,INT,E2E	S

HVPN-P2-205 — Aggregate censorship-evasion telemetry (08–... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	Record aggregate "transport X after N escalations in region R" (no per-user data) + Grafana panel	dashboard renders transport-mix by region	UNIT,CHAL	S
.2	No-log schema-lint mutation: a per-user evasion table FAILs	the planted per-user table FAILs the build (SEC)	SEC	XS

HVPN-P2-206 — Hysteria2/Salamander interop evaluation (D1/D6) (08–... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	A/B benchmark Hysteria2+Salamander vs MASQUE/H3 (goodput under loss, DPI evasion parity)	recorded §20 decision-log row + CSV; rec: keep MASQUE primary, Hysteria2 optional rung	BENCH,E2E	S

3. E21 — DAITA traffic-analysis defense

Adopt maybenot (the engine behind Mullvad DAITA) — do not roll your own.

HVPN-P2-210 — maybenot engine integration (08–... §7; M)

id	Subtask	Acceptance	Tests	Cx
.1	Embed maybenot::Framework as a shaping stage (on_packet pad, tick cover) over WG datagrams	outgoing datagrams pad to uniform sizes; cover packets emit on schedule (capture: size histogram collapses)	UNIT,INT	M
.2	Machines-as-config (data, not code) loading		UNIT,BENCH	S

id	Subtask	Acceptance	Tests	Cx
		a known padding machine loads + drives shaping		

HVPN-P2-211 — Machines-as-data distribution via NetworkMap (08-... §7; S)

id	Subtask	Acceptance	Tests	Cx
.1	NetworkMap.daita field carrying signed maybenot machine defs	a coordinator-pushed machine update changes shaping live (no rebuild)	UNIT,INT,E2E	S
.2	Signature-verify machine blobs before load	a tampered blob is rejected (captured)	SEC,INT	S

HVPN-P2-212 — DAITA opt-in toggle + honest cost surface (08-... §7; S)

id	Subtask	Acceptance	Tests	Cx
.1	DaitaToggle widget (off by default) + Riverpod binding + live cover_ratio readout	UX MP4 shows toggling DAITA, the cost note, the live overhead (§11.4.159 vision-verified)	UI,UX,REC	S

HVPN-P2-213 — Closed-world fingerprinting efficacy harness (the privacy claim) (08-... §7; L)

id	Subtask	Acceptance	Tests	Cx
.1	Closed-world WF classifier trained on captured traces; DAITA-off vs DAITA-on	DAITA-on accuracy < DAITA-off by the documented margin on the project's own fixtures	FA,SEC,BENCH	L
.2	Self-validated analyzer (golden-good reduces accuracy; golden-bad no-op machine does NOT, §11.4.107(10))	the golden-bad no-op machine FAILs the analyzer (proves no-bluff)	CHAL,SEC	M
.3	Cover-traffic overhead recorded	overhead captured + surfaced honestly	BENCH	S

4. E22 — Direct P2P + NAT traversal + relay

The gateway becomes a coordinator + relay-of-last-resort; traffic goes direct where NAT allows.

HVPN-P2-220 — STUN-like endpoint discovery (08-... §8; M)

id	Subtask	Acceptance	Tests	Cx
.1	Host + server-reflexive candidate discovery via the gateway STUN endpoint → ReportStatus	a node behind NAT reports a correct server-reflexive ip:port (INT)	UNIT,INT,E2E	M
.2	STUN endpoint refuses amplification (response ≤ request, rate-limited)	flood test: no amplification, rate-limit holds (DDOS)	DDOS	S

HVPN-P2-221 — Candidate distribution via WatchNetworkMap (08–... §8; S)

id	Subtask	Acceptance	Tests	Cx
.1	Reuse WatchNetworkMap as the signaling channel: push each peer the other's candidates as a Peer.endpoint delta	both peers receive candidates within the SLO1 budget (captured)	UNIT,INT,E2E	S
.2	Candidates policy-filtered (need-to-know)	an unauthorized peer never appears in candidates (SEC)	INT,SEC	S

HVPN-P2-222 — UDP hole punching across the NAT-type matrix (08–... §8; L)

id	Subtask	Acceptance	Tests	Cx
.1	Simultaneous WG handshake probes to all candidates; WG roaming latches the working src	direct datagrams captured bypassing the gateway for full/restricted/port-restricted pairs	UNIT,E2E,STRESS	L
.2	Symmetric-on-both-ends → clean relay fallback	symmetric pairs fall through to relay; no false "direct" claim (§11.4.107)	SEC,CHAL	M

HVPN-P2-223 — helix-relay (DERP-style encrypted relay fallback) (08–... §8; L)

id	Subtask	Acceptance	Tests	Cx
.1	Edge relay mode keyed by destination pubkey (below the WG crypto boundary)	a symmetric-NAT pair communicates via relay (E2E)	UNIT,INT,E2E	L
.2	Relay holds only ciphertext, no per-flow durable record; tenant/region-fungible	memory/pcap inspection proves only pubkey-keyed ciphertext (no plaintext)	SEC,STRESS	M
.3	Relay forwarding throughput	captured forwarding BENCH	BENCH	S

HVPN-P2-224 — Path selection: start-relay, upgrade-to-direct (08-... §8; M · SLO1)

id	Subtask	Acceptance	Tests	Cx
.1	Start on relay, attempt direct in background, upgrade seamlessly; report path	SLO1: direct-path establishment (non-symmetric NAT) > 90% within 10 s (histogram)	UNIT,E2E,PERF,FA	M
.2	No connectivity gap on upgrade	relay is the instant fallback; no user-visible reconnect	FA	S

HVPN-P2-225 — Continuous direct-path health-check + downgrade (08-... §8; S · SLO2)

id	Subtask	Acceptance	Tests	Cx
.1	Health-loop: relay→direct upgrade < 15 s, no reconnect; direct→relay downgrade on degradation	SLO2 < 15 s upgrade (captured); injected degradation downgrades within the window	UNIT,E2E,PERF,CHAOS	S

HVPN-P2-226 — Path indicator UI + UX (08-... §8; S)

id	Subtask	Acceptance	Tests	Cx
.1	PathIndicator bound to the core status stream	UX MP4 shows relay→direct on upgrade and direct→relay on injected degradation (§11.4.159)	UI,UX,REC	S

5. E23 — Multi-hop (nested WireGuard)

Client → Entry → Exit so no single node sees both who you are and where you go.

HVPN-P2-230 — Hop-chain computation in the coordinator (08-... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Compute the hop chain (respect exitNodes + user selection), per-hop keys, hops list in NetworkMap	Entry=EU/Exit=US yields a two-hop map with distinct per-hop keys (INT)	UNIT,INT	M
.2	Enforce exitNodes policy	an unauthorized exit is never offered (SEC)	SEC	S

HVPN-P2-231 — Nested-session construction in helix-core (08–... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Build nested WG (outer Client [?] Entry obfuscated, inner Client [?] Exit plain) — transport/DAITA on outer only	end-to-end reachability through Entry→Exit (E2E)	UNIT,E2E	M
.2	Second-hop overhead measured	goodput/latency overhead captured + documented	BENCH	S

HVPN-P2-232 — Hop-isolation capture assertions (the privacy proof) (08–... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Pcap at each hop: Entry sees client IP not dest; Exit sees dest not client IP	P2-AC4 both isolation invariants proven by pcap	SEC,E2E,CHAL	M
.2	Mutation: collapse to single hop FAILs the assertion	the single-hop-collapse mutation FAILs	CHAL,SEC	S

HVPN-P2-233 — Multi-hop picker UI (08–... §9; S)

id	Subtask	Acceptance	Tests	Cx
.1	MultiHopPicker (region/jurisdiction labelled) bound to the policy-permitted hop set	UX MP4 shows selecting Entry/Exit + the two-hop connection with jurisdiction labels (§11.4.159)	UI,UX,REC	S

6. E24 — Post-quantum handshake (ML-KEM PSK)

Close harvest-now-decrypt-later via a PQ-derived PSK mixed into WG — without forking WG crypto.

HVPN-P2-240 — ML-KEM PSK exchange over the control channel (08–... §10; L)

id	Subtask	Acceptance	Tests	Cx
.1	ML-KEM (FIPS 203) exchange over the authenticated Coordinator RPC (no new listener) → HKDF → WG PSK	two PQ-capable peers establish a WG session whose PSK is ML-KEM-derived (captured)	UNIT,INT,SEC	L
.2	PSK rotates on each rekey; exchange at setup/rekey only	PSK rotation captured; no per-packet KEM	SEC,BENCH	M

HVPN-P2-241 — Hybrid combiner (never PQ-only) + capability negotiation (08–... §10; M)

id	Subtask	Acceptance	Tests	Cx
.1	PSK = HKDF(classical pq) combiner + capability flags	PQ-capable[?]classical-only pairs interoperate (negotiate down without error)	UNIT,SEC,E2E	M
.2	Mutation: PQ-only combine FAILs	a PQ-only combine FAILs the unit + mutation gate	UNIT,SEC	S

HVPN-P2-242 — Downgrade-safety guard (08–... §10; M)

id	Subtask	Acceptance	Tests	Cx
.1	Downgrade FSM: a PQ failure → classical-or-fail-closed, never weaker	injecting a PQ failure yields classical-or-fail-closed, never weaker-than-classical (captured)	SEC,E2E,CHAL	M
.2	MITM strip-PQ detection logged as a control event	a strip-PQ attempt is detected/logged (not as traffic)	SEC	S

HVPN-P2-243 — PQ default-on where supported + adoption metric (08–... §10; S · SLO5)

id	Subtask	Acceptance	Tests	Cx
.1	PQ default-on for capable pairs + PqChip + helix_pq_sessions_ratio	SLO5: PQ adoption among capable pairs > 95% (dashboard)	UNIT,PERF,REC	S
.2	Zero steady-state per-packet cost	per-packet benchmark ≈ 0 delta vs classical	PERF	S

HVPN-P2-244 — Rosenpass alternative evaluation (decision) (08–... §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	Evaluate Rosenpass vs in-house ML-KEM (audit pedigree, integration cost)	recorded §20 decision row; rec: ship in-house ML-KEM, keep a Rosenpass KemSuite adapter	BENCH,SEC	S

7. E25 — Desktop apps (Windows, macOS)

Everything above the shim is the same helix-ui + helix-core.

HVPN-P2-250 — Windows privileged service + wireguard-nt shim (08... §11; L)

id	Subtask	Acceptance	Tests	Cx
.1	Windows service hosting helix-core over wireguard-nt/wintun + named-pipe IPC	connect → reach an authorized LAN host from Windows; app unprivileged, service holds tunnel	INT,E2E,REC	L
.2	Code-signed driver + service (Authenticode)	driver + service signature verified	SEC	M

HVPN-P2-251 — macOS NEPacketTunnelProvider shim (08... §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	macOS NE hosting helix-core (desktop memory budget), notarized + signed	connect → reach from macOS; notarization + signature verified; UX recording (§11.4.159)	INT,E2E,SEC,REC	M

HVPN-P2-252 — Desktop split-tunnel (per-app routing) (08... §11; L)

id	Subtask	Acceptance	Tests	Cx
.1	Windows WFP filters + macOS app-bound rules, include/exclude set from the shared UI	a designated app routes outside the tunnel while the rest stays inside (pcap proves the split per-app on both OSes)	UNIT,E2E,SEC	L
.2	Split-tunnel UX recording	UX MP4 of the split-tunnel screen (§11.4.159)	UX,REC	S

HVPN-P2-253 — Desktop kill-switch + DNS-leak parity (08... §11; S)

id	Subtask	Acceptance	Tests	Cx
.1	WFP (Win) + NE rules (mac) kill-switch wired to core state	on forced drop, host-side pcap shows zero plaintext + zero DNS leak on both OSes (§11.4.107)	SEC,E2E,REC	S

8. E26 — Policy-as-code & GitOps

Git becomes the source of truth; the apply runner is the tenant's own runner (§11.4.156, no active server-side CI).

HVPN-P2-260 — helixvpncctl policy compile --dry-run effect-diff (08... §12; M)

id	Subtask	Acceptance	Tests	Cx
.1	Render the compiled <i>effect</i> delta (which devices gain/lose which access)	a PR adding one ACL line produces a human-readable effect-diff naming the exact devices/targets	UNIT,INT	M
.2	Mutation: a hidden granted edge FAILs the diff completeness check	the hidden-edge mutation FAILs	SEC,CHAL	S

HVPN-P2-261 — GitOps apply runner + tenant-token auth (08–... §12; M · P2-AC7)

id	Subtask	Acceptance	Tests	Cx
.1	On merge, tenant runner helixvpctl policy apply (token-auth) → persist version → emit policy.compiled → converge	P2-AC7: a merged PR converges the fleet < 1 s (helix_reconcile_seconds p99) with the effect-diff reviewed	INT,E2E,FA	M
.2	No secrets in the repo (declarative intent only; sops/age refs)	secret-leak audit green (§11.4.10)	SEC	S

HVPN-P2-262 — GitOps round-trip safety (drift detect + rollback) (08–... §12; S)

id	Subtask	Acceptance	Tests	Cx
.1	helixvpctl policy diff/ rollback; idempotent apply	an out-of-band change is flagged as drift; a revert PR restores the prior compiled state byte-identically (captured)	INT,CHAOS,SEC	S

9. E27 — HA + multi-region fleet

Same images, only topology changes (08–... §13/§17).

HVPN-P2-270 — Stateless coordinator fleet + stream re-pin (08–... §13; L)

id	Subtask	Acceptance	Tests	Cx
.1	N stateless coordinators per region; rebuildable cache from PG+NATS	coordinator holds no local durable state (§11.4.108 clean-restart)	UNIT,INT,SCALE	M
.2			CHAOS	M

id	Subtask	Acceptance	Tests	Cx
	Stream re-pin: agent re-pins to a sibling, resumes from known_version	killing a coordinator mid-stream re-pins + resumes with zero missed deltas (captured)		

HVPN-P2-271 — Postgres HA (Patroni) + regional read replicas (08-... §13; L)

id	Subtask	Acceptance	Tests	Cx
.1	Single logical primary + sync replicas + Patroni auto-failover	killing the PG primary fails over with no committed-write loss (CHAOS)	INT,CHAOS,SCALE	L
.2	Regional read replicas serving coordinator hydration; RLS/no-log on replicas	regional coordinators hydrate from local replicas; invariants hold (SEC)	SEC,SCALE	M

HVPN-P2-272 — NATS JetStream cross-region event mesh (Redis swap) (08-... §13; L · SLO4)

id	Subtask	Acceptance	Tests	Cx
.1	NatsJetStreamBus impl behind the unchanged EventBus interface (durable subjects = Phase-1 taxonomy 1:1)	the taxonomy is identical to Phase 1 (no event renamed)	UNIT,INT	M
.2	Cross-region propagation p99 < 2 s	SLO4 cross-region p99 < 2 s (captured)	E2E,PERF,SCALE	M
.3	Region partition + heal shows no event loss	durable subjects replay; no loss (CHAOS)	CHAOS	M

HVPN-P2-273 — Edge/relay fleet + client steering (Anycast/GeoDNS + RTT list) (08-... §13; M)

id	Subtask	Acceptance	Tests	Cx
.1	Anycast + GeoDNS fallback; NetworkMap.gateways candidate list; client picks by RTT	a client picks the nearest edge by measured RTT; Anycast routes UDP/QUIC to the closest PoP	INT,E2E,PERF	M
.2	STUN/edge handshake flood rate-limited	flood shed without dropping legit tunnels (DDOS)	DDOS	S

HVPN-P2-274 — Regional failover flow + floating IPs (08-... §13; M · P2-AC8/SLO3)

id	Subtask	Acceptance	Tests	Cx
.1	Health prober → gateway.failover{from,to} → NetworkMap delta → client re-handshake; floating IPs	SLO3: gateway failover → client reconnected p99 < 3 s P2-AC8: region kill reconnects clients < 3 s,	E2E,CHAOS,PERF	M
.2	Existing direct-P2P sessions unaffected by region loss	zero policy/identity loss, direct sessions survive (chaos capture)	CHAOS	M

10. E28 — Observability additions

Counts + health only, never destinations/flows (08-... §14).

HVPN-P2-280 — Reach/evasion KPI dashboards (08-... §14; M)

id	Subtask	Acceptance	Tests	Cx
.1	Transport-mix/region, escalation depth, direct-vs-relay ratio, hole-punch success by NAT-type, PQ adoption %, DAITA overhead % (aggregate)	dashboards render every KPI from aggregate counters	INT,CHAL	M
.2	No-log lint mutation: a per-flow source table FAILs	the per-flow mutation FAILs the no-log lint (§11.4 / §02 §2.4)	SEC	S

HVPN-P2-281 — Multi-region health (NATS lag, PG replication, convergence p99) (08-... §14; S)

id	Subtask	Acceptance	Tests	Cx
.1	Per-region coordinator health + NATS consumer lag + PG replication lag + cross-region convergence-p99 panels + alerts	convergence-p99 panel reflects P2-SLO4 + alerts on breach; lag alerts fire under an injected partition (captured)	INT,PERF,SCALE	S

11. E29 — QA · SLOs · DoD certification

Each surface is a measure-don't-assert gate; privacy/efficacy claims carry self-validated analyzers (§11.4.107(10)).

Parent task	Subtask focus (. 1/. 2)	Acceptance	Tests	Cx
290 NAT-traversal matrix	. 1 full/restricted/port-restricted/symmetric harness	every theoretically-direct pair achieves direct (pcap bypass); symmetric relays cleanly; no false direct	E2E,STRESS,FA,CHAL	L
291 obfuscation gauntlet	. 1 each transport through its DPI block + tshark classifier	P2-AC1: each transport survives its block + the classifier confirms the disguise P2-AC2:	E2E,SEC,CHAL	M
292 DAITA efficacy cert	. 1 accuracy-drop with the self-validated analyzer wired into meta-test	DAITA-on accuracy drops materially; golden-bad no-op FAILs the analyzer; cost surfaced	FA,SEC,CHAL	M
293 PQ interop + downgrade cert	. 1 classical [?] PQ negotiation + downgrade safety	P2-AC5: all capability pairs interoperate; every injected PQ failure → classical-or-fail-closed (captured)	SEC,E2E,CHAL	M
294 multi-hop isolation cert	. 1 per-hop pcap assertions	P2-AC4: both isolation invariants proven; single-hop-collapse mutation FAILs	SEC,E2E,CHAL	S
295 HA/chaos cert	. 1 kill coordinator/PG primary/region	P2-AC8 + SLO3 (failover p99<3s) + SLO4 (cross-region	CHAOS,SCALE,PERF,FA	L

Parent task	Subtask focus (.1/.2)	Acceptance	Tests	Cx
296 DDoS resilience cert	.1 STUN + edge handshake flood	region p99<2s) green; no control-plane data loss; direct sessions survive STUN response ≤ request (no amplification); edge sheds a flood without dropping legit tunnels below SLO	DDOS,SEC,STRESS	M
297 Phase-2 DoD cert + discovery sweep	.1 full-suite retest + enumerated discovery (§11.4.118)	all 8 P2-AC + 5 P2-SLO green with evidence; the discovery set surfaces zero release-blockers (else §11.4.129)	FA,CHAL,REC	M

12. E30 — Governance & release

HVPN-P2-300 — Phase-2 items DB projection + docs-chain sync (08-... §16; M)

id	Subtask	Acceptance	Tests	Cx
.1	Project the Phase-2 leaves into docs/workable_items.db via the reused Phase-1 loader + .docs_chain/contexts/wbs-p2.yaml	md ² db byte-identical round-trip (§11.4.93); a leaf edit re-syncs exports; verify is the gate	UNIT,INT,FA	M

HVPN-P2-301 — Phase-2 release tagging + multi-upstream publish (08-... §16; S)

id	Subtask	Acceptance	Tests	Cx
.1	Cut <HELIX_RELEASE_PREFIX>-2.0.0-parity (§11.4.151) on main + every owned submodule; publish via merge-onto-latest-main (no force-push §11.4.113)	git tag -l '<PREFIX>-2.*' enumerates the whole release; HVPN-P2-297 cert is the tag's evidence;	FA	S

id	Subtask	Acceptance	Tests	Cx
		reachable on every remote		

13. Subtask roll-up + measure-don't-assert note

Epic	Parent tasks	Deepened subtasks	Risk note
E19 entry seams	3	3	prereq audits
E20 transport set	7	15	additive impls of one trait
E21 DAITA	4	8	measure-don't-assert
E22 P2P/NAT	7	14	no-false-direct (§11.4.107)
E23 multi-hop	4	7	isolation proven by pcap
E24 PQ	5	9	never-weaker-than-classical
E25 desktop	4	6	per-app split-tunnel
E26 GitOps	3	5	effect-diff completeness
E27 HA/multi-region	5	11	failover SLOs
E28 observability	2	4	aggregate-only, no-log holds
E29 QA/DoD	8	8	self-validated analyzers
E30 governance	2	2	items-DB + tag
Total	54	92	—

The two privacy-claim epics (E21 DAITA, E24 PQ) and the NAT-traversal epic (E22) carry the load-bearing anti-bluff escalations (§11.4.107): a green pass that *asserts* a privacy claim without the measurement is a §11.4 PASS-bluff; a "direct" path claimed without a non-gateway src observed at the peer is the same class. Every subtask's acceptance is a captured-evidence assertion (§11.4.5/.69/.107); the golden-bad fixture FAILing the analyzer is the proof the harness cannot bluff (§11.4.107(10)). Complexity sums to $\approx 08\text{--}... \text{ §19's } \sim 259$ person-days — sizing only, never a date (§11.4.6).

Sources verified

- 08-phase2-parity-wbs.md §5–§16 (every task Desc/Deliverable/Acceptance/Effort/Tests), §17 HA topology, §18 traceability, §19 critical path, §20 open decisions — read 2026-06-26.
- Sibling workable-items-model.md (§3/§6/§9), dependency-graph.md (§5 Phase-2 DAG), subtask-deepening-p1.md (§0 conventions) — authored this volume.

- Constitution anchors

§11.4.5/.6/.10/.40/.54/.58/.69/.85/.91/.93/.107/.108/.113/.118/.129/.132/.151/.156/.159/.169 — read 2026-06-26.

Honest boundary (§11.4.6): subtasks decompose stated Phase-2 work.
DAITA/PQ efficacy subtasks are *measure-don't-assert* — their acceptance is a measured delta with a self-validated analyzer, never an asserted claim.
Complexity is sizing TARGET only; DDOS subtasks are now applicable (public multi-region surface), the Phase-1 NOT_APPLICABLE deferral re-armed.